

ADVENTURE WORKS

- PERFORMANCE INSIGHTS -

BY GROUP 1

MEMBERS

NIKITHA

AMITH RAJ

VAITHEESWARAN

YAMJALA SREESHANT

RUCHA SIDDHARTH POKALE

DEEPALI RAJKUMAR SUDAKE

INTRODUCTION

Project Overview

- > Multi-tool analytics project covering sales data from Adventure Works
- > Data integrated from Fact InternetSales & Fact Internet Sales New (Union)
- > Enriched with Product, Customer & Territory lookups
- > Spanning years 2010-2014 with 60,000+ order records

Objectives

- > Build unified sales dataset with calculated KPIS
- > Analyse performance by year, month, quarter & region
- > Create interactive dashboards in Excel, Power BI & Tableau
- > Deliver actionable insights for management decisions

Dataset Details

- > Source: Adventure Works Data Warehouse
- > Tables: Fact Sales, Product, Customer, Territory
- > Key fields: Order Date, Unit Price, Cost, Quantity
- > Calculated: Sales Amount, Production Cost, Profit, Date fields

KEY PERFORMANCE INDICATORS - OVERVIEW



Total Sales Amount

₹29.36M

Across all regions & years



Total Production Cost

₹17.28M

Manufacturing & production



Total Profit

₹12.08M

Net profit (Sales – Cost)



Total Orders

60,398

Customer transactions



Max Single Sale

₹3,578

Highest single order value



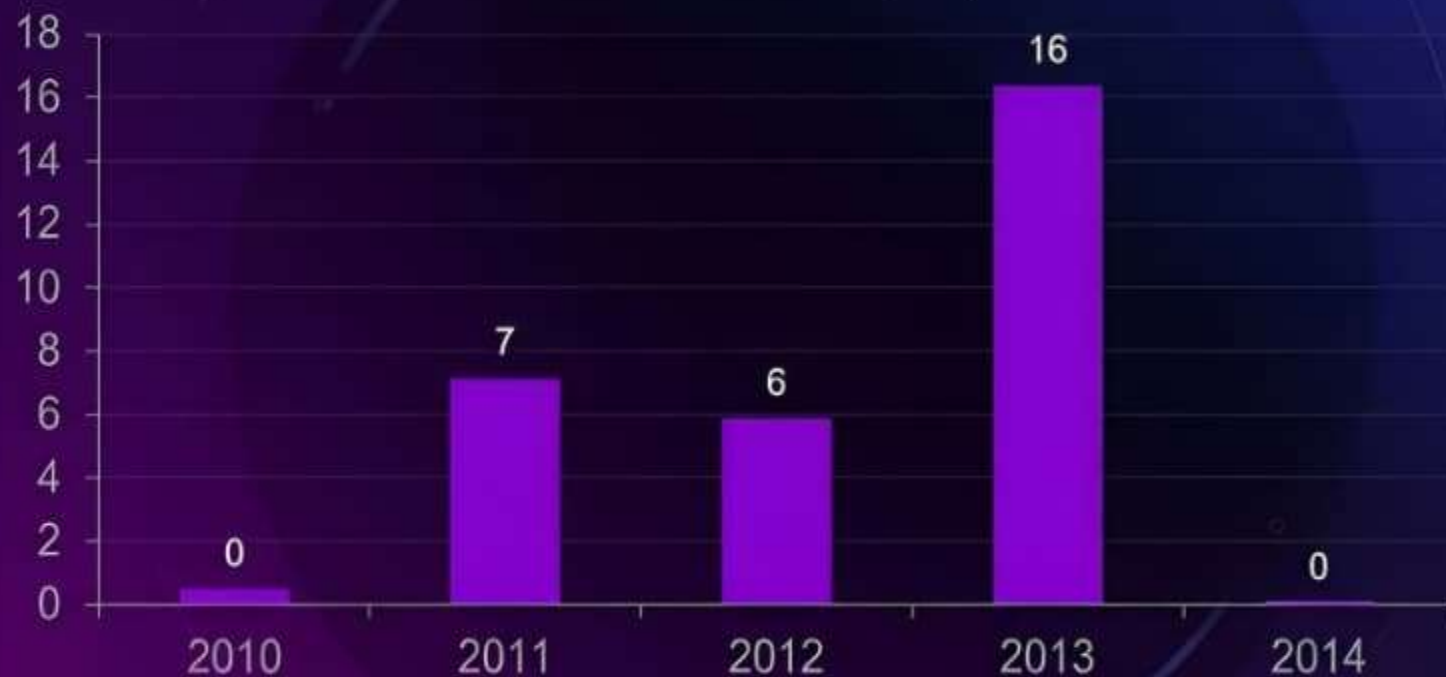
Profit Margin

40.7%

Overall margin on sales

KPI 1 — YEAR-WISE SALES & REVENUE

Year-Wise Sales (₹M)



Definition & Business Impact

Definition:

Total revenue generated = Unit Price × Quantity – Discount

Why It Matters:

Primary revenue indicator for growth tracking & forecasting

Key Findings:

- 2013 was peak year (₹16.35M — 56% of total)
- Strong growth 2010→2013
- 2014 partial year data
- Bikes dominate at 95% share

2013

Peak Year

₹16.35M Revenue

+130%

YoY Growth

2011 → 2013

Bikes

Top Category

₹28M total

₹7.3M

Avg Per Year

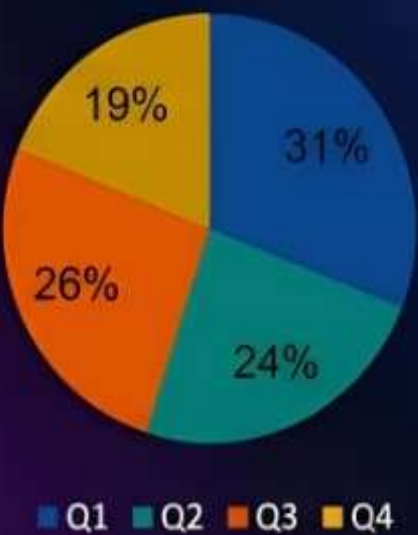
Across 4 years

KPI 2 — MONTH-WISE & QUARTER-WISE SALES

Month-Wise Sales Trend (₹K)



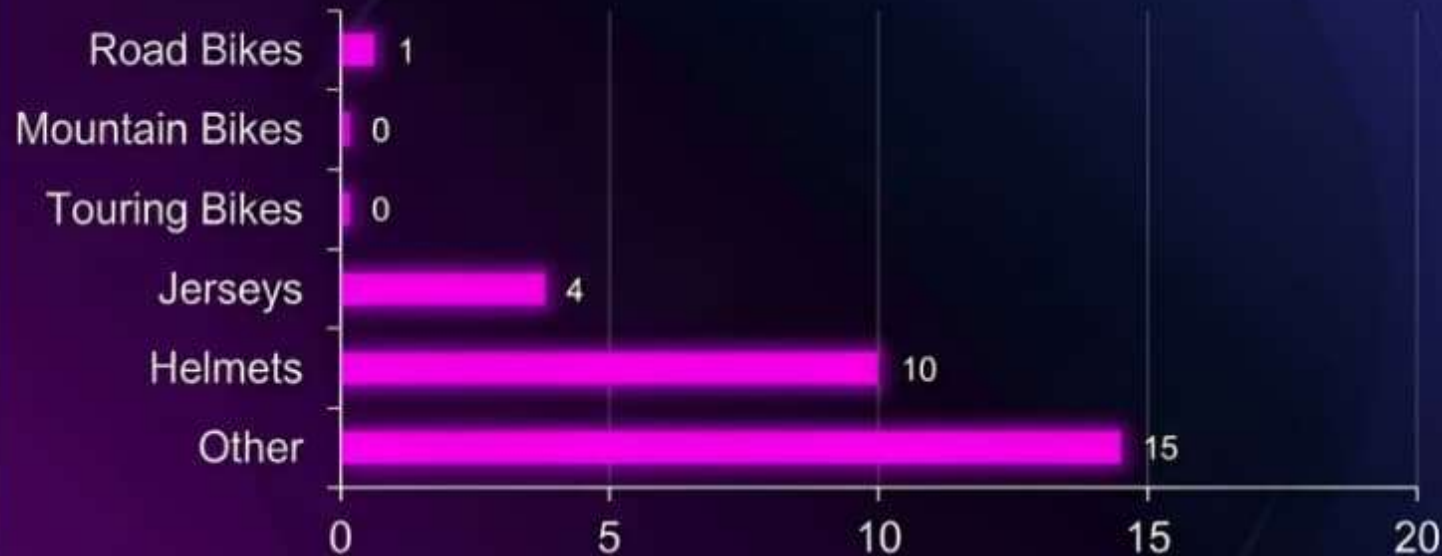
Quarter-Wise Sales Share



Period	Finding	Value	Insight
Month-Wise	Peak sales month	December (₹3,211K)	Holiday season drives highest revenue
Month-Wise	Lowest month	January (₹1,869K)	Post-holiday dip; marketing opportunity
Quarter	Top quarter (Q1=31%)	Q1 dominates	Strong Jan-Mar cycle despite Jan dip
Quarter	Quarter spread	19%–31%	Balanced distribution; Q4 slightly weaker

KPI 3 — PRODUCT & CUSTOMER PERFORMANCE

Product-Wise Sales (₹M)



₹3.56M

Mountain Bikes

P1210 segment

₹9.28M

Road Bikes

P1210 segment

4,000+

Total Customers

Unique buyers

₹486

Avg Order Value

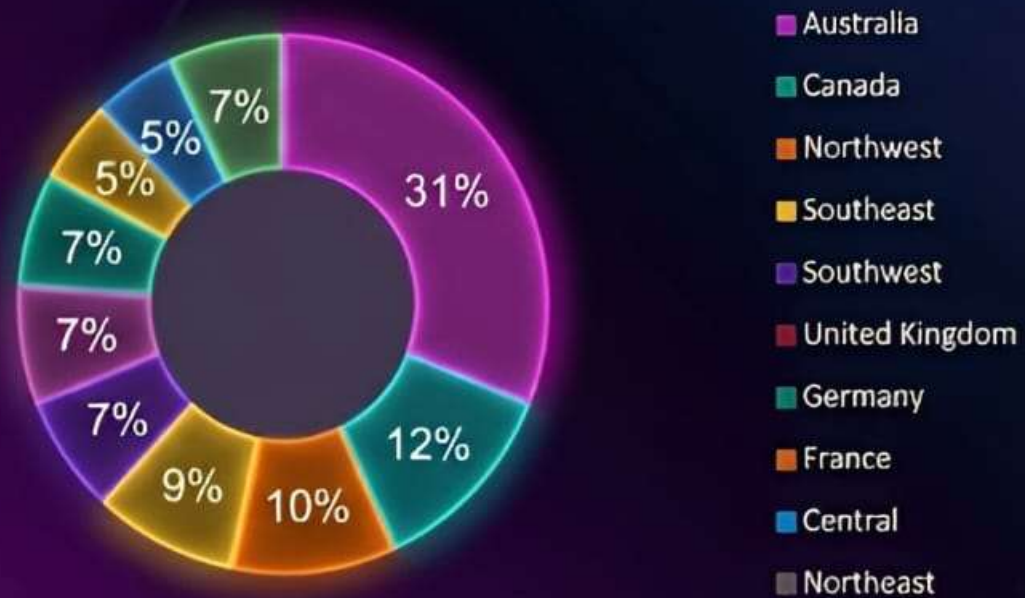
Per transaction

Key Findings & Business Impact

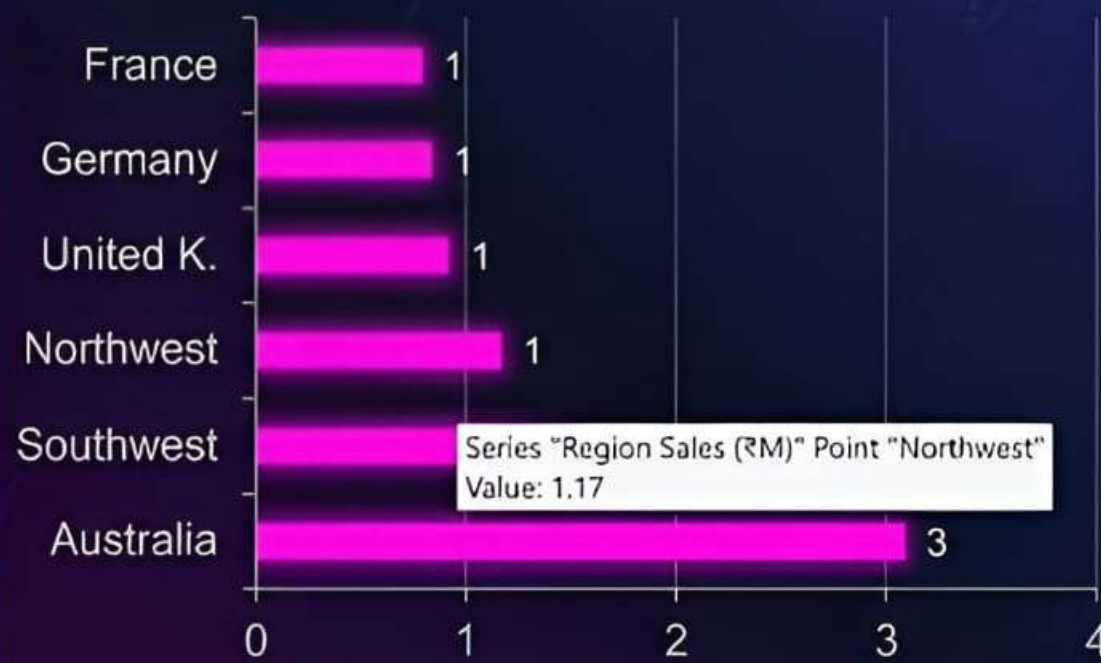
- Road Bikes generate the highest revenue (₹14.5M — 49% of total), followed by Mountain Bikes (₹10M — 34%)
- Accessories & Clothing contribute <5% — opportunity for bundling or upselling strategies
- High-value customers drive disproportionate revenue — customer segmentation can optimize marketing spend
- Product concentration risk: 83% revenue from 2 sub-categories → diversification recommended

KPI 4 — REGION-WISE SALES & PROFIT ANALYSIS

Region-Wise Production Share



Top Regions by Sales (₹M)



Australia

Top Region

31% of production

Australia

Sales Leader

₹3.09M revenue

40.7%

Profit Margin

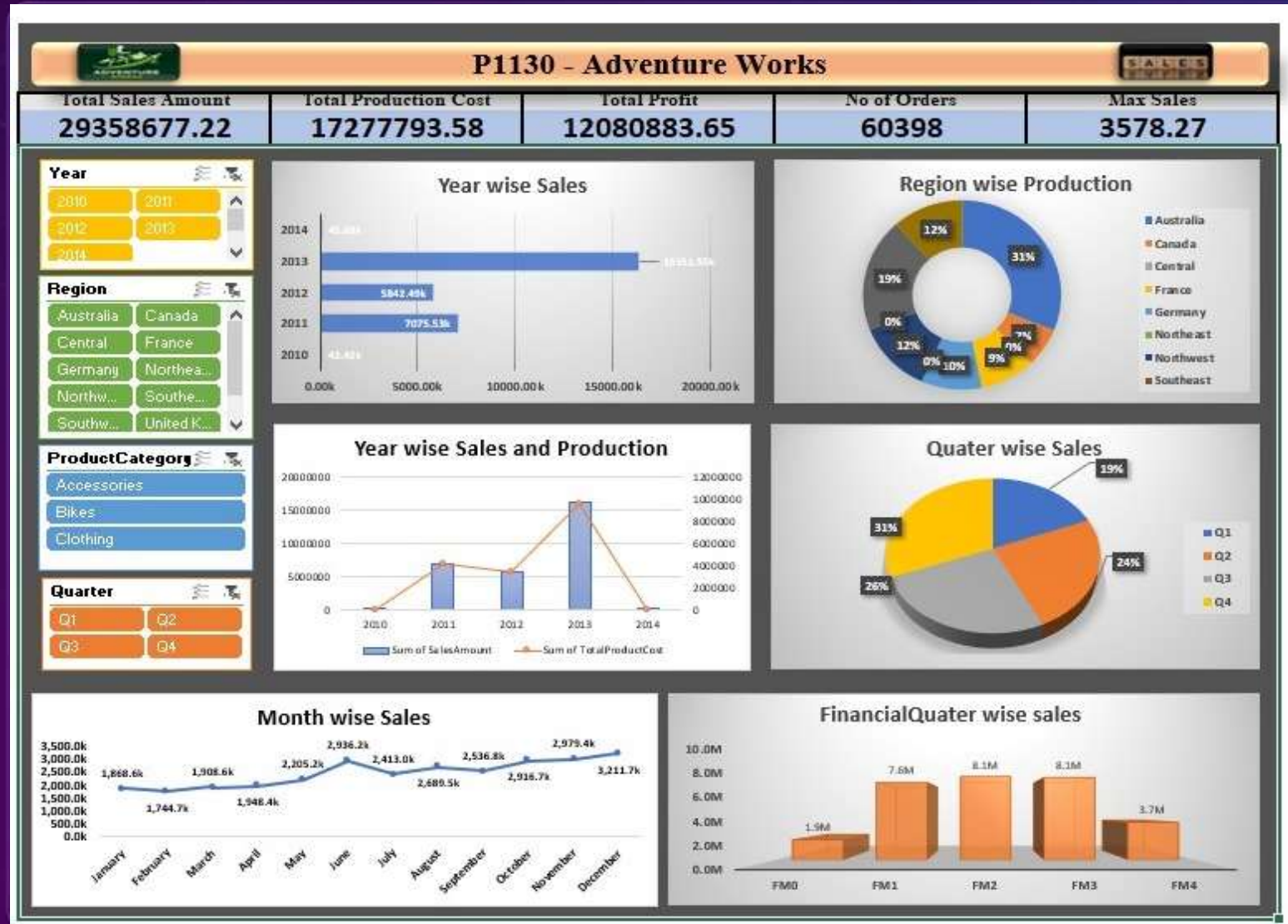
Company average

Southwest

Growth Region

Strong 2011–2013

Power BI Dashboard — Sales & Performance Insights



Dashboard Overview:

- Interactive Power BI dashboard with dynamic filters
- KPIs: Sales, Profit, Cost, Orders

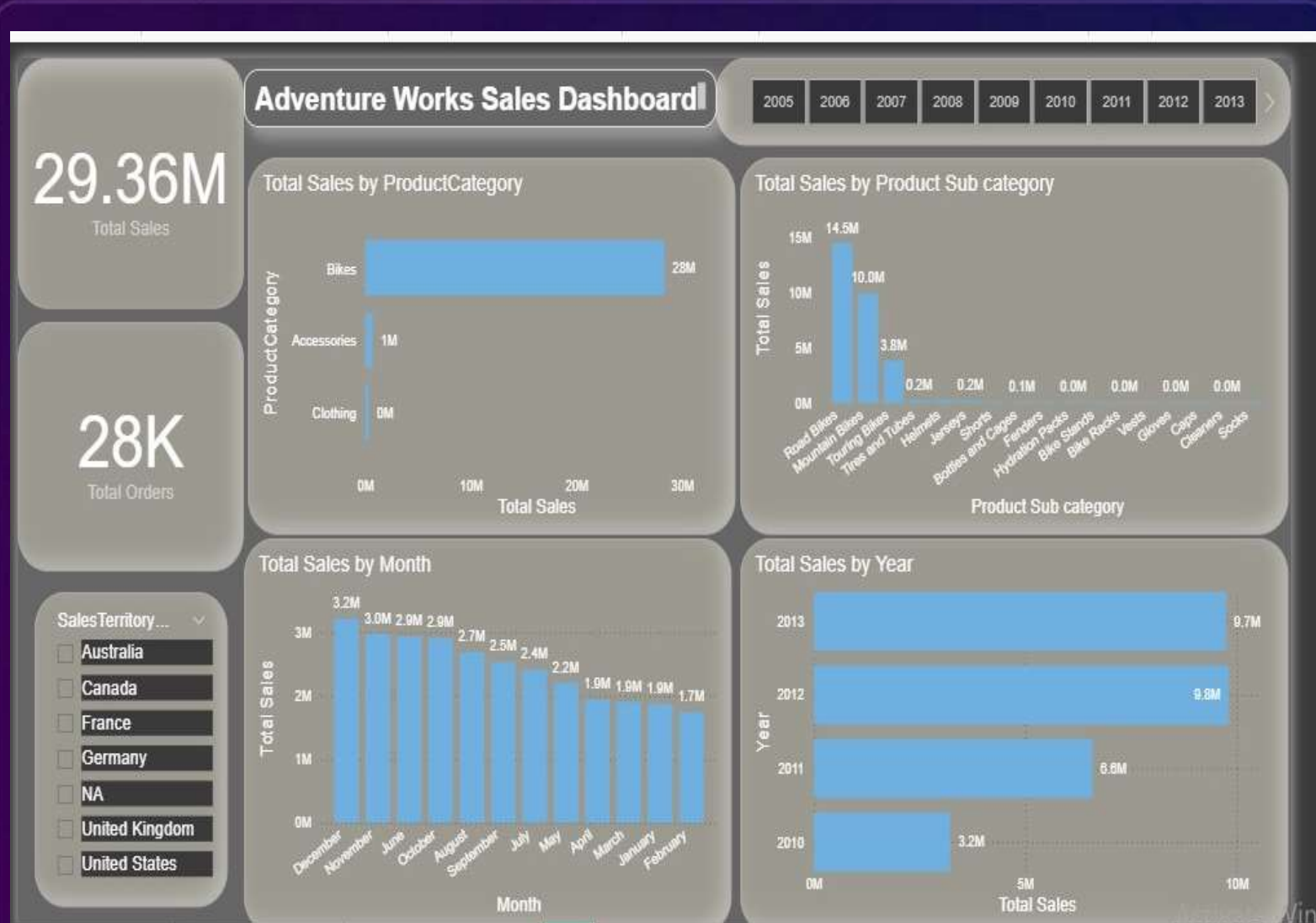
Key Findings:

- 2013 contributed 56% revenue (~₹16.35M)
- FM1–FM3 peak revenue periods
- Bikes dominate >90% sales

Business Impact:

- Faster reporting & self-service insights
- Improved decision-making

Excel Dashboard — Data Processing & KPI Analysis



Dashboard Overview:

- Built using Pivot Tables & Charts
- Integrated multi-source data

Key Findings:

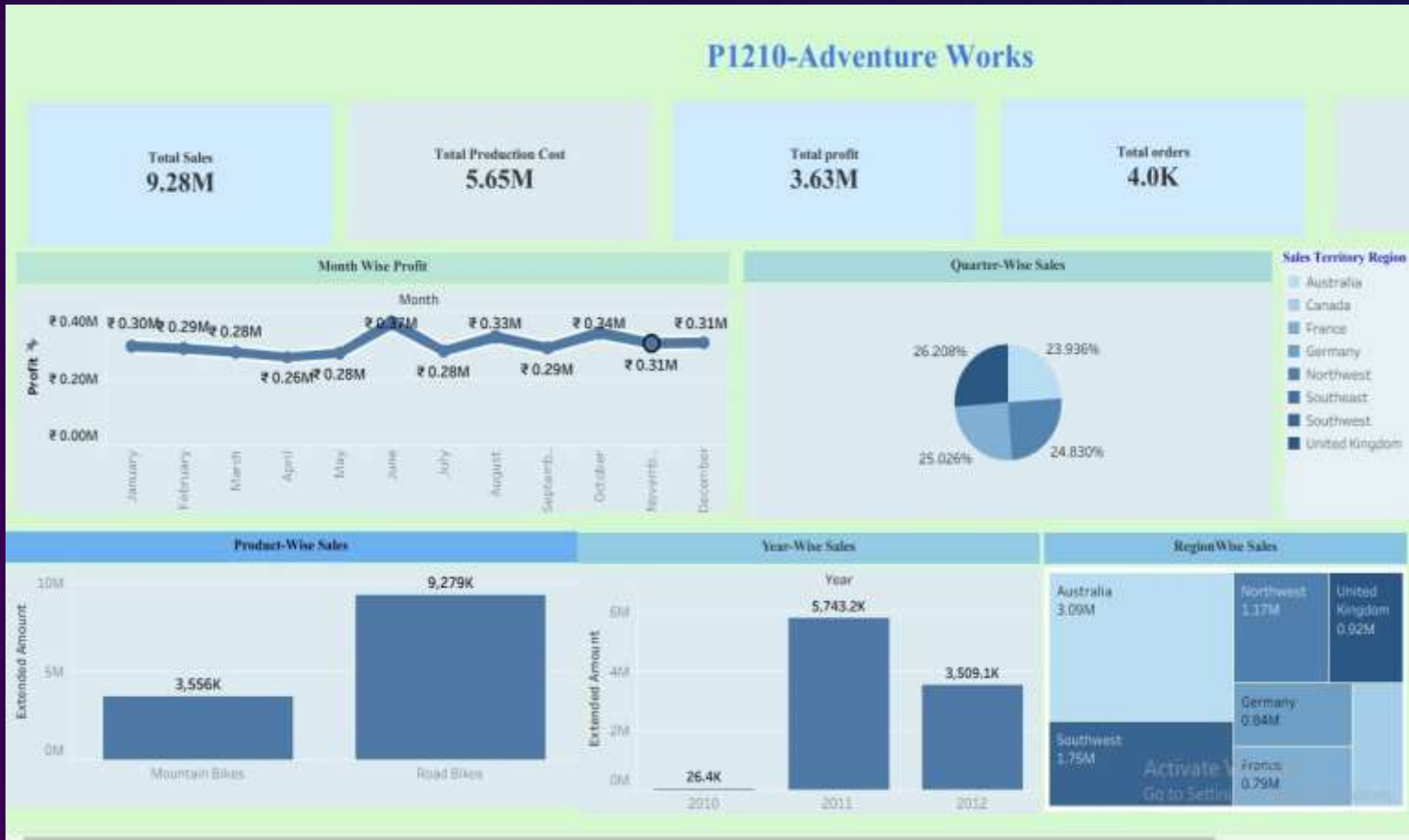
- Total Sales ₹29.36M
- December peak, January lowest
- Cost ~60% of sales

Business Impact:

- Simplified reporting
- Foundation for BI dashboards

Tableau Dashboard — Advanced Visual Analytics

P1210-Adventure Works



Dashboard Overview:

- Advanced visual analytics using Tableau

Key Findings:

- Profit peak in January
- Australia top region ₹3.09M
- Road Bikes 2.6x higher sales

Business Impact:

- Identified high-value regions
- Improved pricing & inventory decisions

BUSINESS IMPACT & CHALLENGES FACED

Business Impact



Revenue Optimization

Identified 2013 as peak year — seasonal strategy can replicate ₹16M+ annual revenue



Customer Targeting

Region-level segmentation enables personalised campaigns; Australia & Southwest are prime markets



Product Strategy

Road Bikes account for 49% revenue — supply chain & inventory must be prioritised accordingly



Cost Control

Production cost runs at ~59% of sales — focus on cost reduction can boost profit margin above 45%



Market Expansion

France & Germany show low penetration relative to revenue potential — growth opportunity

Challenges Faced



Data Challenges

- Union of two Fact tables required careful schema alignment
- OrderDateKey needed transformation into 9 date sub-fields



Technical Challenges

- VLOOKUP across large datasets caused performance issues in Excel
- Tableau relationship mapping required multiple joins & blending



Business Challenges

- Interpreting Financial Quarter logic (FM0–FM4) vs Calendar Quarter
- Defining Sales Amount formula: $(\text{Unit Price} \times \text{Qty}) - (\text{Unit Discount} \times \text{Qty})$

SUMMARY, FINDINGS & CONCLUSION

This project demonstrates a complete data analytics lifecycle from raw data ingestion to interactive dashboard delivery across 3 industry-standard tools.



Highest Revenue Year

2013 with ₹16.35M — driven by Bikes product surge across all regions



Dominant Product

Road Bikes ₹14.5M (49%) — core revenue engine requiring supply priority



Top Region

Australia leads at 31% production share and ₹3.09M in P1210 segment



Seasonal Pattern

Dec peak (₹3.2M), Jan dip (₹1.9M) — marketing campaigns should align



Profit Insights

₹12.08M profit at 40.7% margin — cost optimization can push this to 45%+



Growth Opportunity

France & Germany underserved — expansion investments can unlock ₹2M+ potential

Recommendations & Next Steps:

Invest in Road Bikes supply chain • Target Aus & Southwest for premium campaigns • Expand to FR & DE markets • Build real-time Power BI pipeline • Implement customer loyalty scoring

THANK YOU...

BY GROUP 1